10.6 Emulsions

Emulsions are colloidal solution in which a liquid is dispersed in an another liquid.

Generally there are two types of emulsions.

(i) Oil in water (O/W) **(ii) Water in oil (W/O)**

Example:

Milk is example of the oil in water type emulsion.

Stiff greases are emulsion of water in oil i.e. water dispersed in lubricating oil.

The process of preparation of emulsion by the dispersal of one liquid in another liquid is called Emulsification.

A colloid mill can be used as a homogeniser to mix the two liquid. To have a stable emulsion a small amount of emulsifier or emulsification agent is added.

Several types of emulsifiers are known.

i. Most of the lyophillic colloids also act as emulsifiers. Example: glue, gelatine.
ii. Long chain compounds with polar groups like soap and sulphonic acids.
iii. Insoluble powders like clay and lamp black also act as emulsifiers.

Identification of types of emulsion

The two types of emulsions can be identified by the following tests.

(i) Dye test:

A small amount of dye soluble in oil is added to the emulsion. The emulsion is shaken well. The aqueous emulsion will not take the colour whereas oily emulsion will take up the colour of the dye.

(ii)Viscosity test

Viscosity of the emulsion is determined by experiments. Oily emulsions will have higher value than aqueous emulsion.

(iii) Conductivity test

Conductivity of aqueous emulsions are always higher than oily emulsions.

(iv) Spreading test

Oily emulsions spread readily than aqueous emulsion when spread on an oily surface.

10.5.1 Deemulsification:

Emulsion can be separated into two separate layers. The process is called Deemulsification.

Various deemulsification techniques are given below

1. Distilling of one component
2. Adding an electrolyte to destroy the charge.
3. Destroying the emulsifier using chemical methods.

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